



Used Car Market Dataset Exploratory Data Analysis (EDA)

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Introduction

This project performs Exploratory Data Analysis (EDA) on a used car market dataset.

The dataset contains information about car brands, fuel types, prices, mileage, transmission types, and cities.

```
In [2]: df = pd.read_csv("/content/final.csv")
```

Dataset Overview

The dataset helps analyze different factors affecting used car prices and market trends.

What are the first five rows of the dataset?

```
In [5]: df.head()
```

```
Out[5]:
```

	brand	model	year	fuel_type	transmission	km_driven	owner_type	city
0	Honda	Amaze	2025	Petrol	Manual	11067	Third	Delhi
1	Tata	Altroz	2025	Diesel	Automatic	9114	First	Delhi
2	Tata	Punch	2021	CNG	Automatic	75049	Second	Delhi
3	Maruti	Brezza	2021	Petrol	Manual	74402	Second	Mumbai
4	Mahindra	Scorpio	2022	Diesel	Automatic	38984	Second	Delhi

5 rows × 23 columns

What are the last five rows of the dataset?

```
In [9]: df.tail()
```

Out[9]:

	brand	model	year	fuel_type	transmission	km_driven	owner_type	
195	Tata	Nexon	2025	Petrol	Automatic	8131	Third	
196	Renault	Triber	2022	Petrol	Automatic	51263	First	
197	Kia	Sonet	2023	Petrol	Manual	29035	Second	
198	Toyota	Fortuner	2019	Diesel	Manual	112378	Third	Na
199	Tata	Nexon	2018	Petrol	Manual	116241	First	Mu

5 rows × 23 columns

What is the shape of the dataset?

In [7]: `df.shape`

Out[7]: (200, 23)

What is the basic information of the dataset?

In [8]: `df.info()`

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 23 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   brand                                200 non-null    object
1   model                                200 non-null    object
2   year                                  200 non-null    int64
3   fuel_type                            200 non-null    object
4   transmission                          200 non-null    object
5   km_driven                            200 non-null    int64
6   owner_type                           200 non-null    object
7   city                                  200 non-null    object
8   mileage                              200 non-null    int64
9   engine_cc                            200 non-null    int64
10  condition_rating                     200 non-null    int64
11  original_price                       200 non-null    int64
12  listed_price                         200 non-null    int64
13  estimated_final_price                200 non-null    int64
14  days_to_sell                        200 non-null    int64
15  bargain_percent                     200 non-null    int64
16  car_age                             200 non-null    int64
17  depreciation_amount                 200 non-null    int64
18  resale_value_percent                 200 non-null    float64
19  demand_score                        200 non-null    float64
20  brand_popularity_score               200 non-null    int64
21  price_per_km                        200 non-null    float64
22  premium_car                         200 non-null    int64
dtypes: float64(3), int64(14), object(6)
memory usage: 36.1+ KB

```

What is the statistical summary of the dataset?

In [10]: `df.describe()`

Out[10]:

	year	km_driven	mileage	engine_cc	condition_rating	original_price
count	200.000000	200.000000	200.000000	200.000000	200.000000	200.000000
mean	2020.785000	63461.430000	18.515000	1417.000000	6.975000	1417000.000000
std	3.622761	47226.542849	4.134640	433.747408	1.485060	1417000.000000
min	2010.000000	8131.000000	12.000000	999.000000	5.000000	500000.000000
25%	2019.000000	29013.000000	15.000000	1197.000000	5.000000	890000.000000
50%	2022.000000	50760.000000	19.000000	1199.000000	7.000000	1000000.000000
75%	2024.000000	84226.000000	22.000000	1497.000000	8.000000	1500000.000000
max	2025.000000	232796.000000	25.000000	2755.000000	10.000000	4600000.000000

Are there any missing values in the dataset?

```
In [11]: df.isnull().sum()
```

```
Out[11]:
```

	0
brand	0
model	0
year	0
fuel_type	0
transmission	0
km_driven	0
owner_type	0
city	0
mileage	0
engine_cc	0
condition_rating	0
original_price	0
listed_price	0
estimated_final_price	0
days_to_sell	0
bargain_percent	0
car_age	0
depreciation_amount	0
resale_value_percent	0
demand_score	0
brand_popularity_score	0
price_per_km	0
premium_car	0

dtype: int64

Car Brand Analysis

Which car brand is most common in the dataset?

```
In [12]: df["brand"].value_counts()
```

```
Out[12]:
```

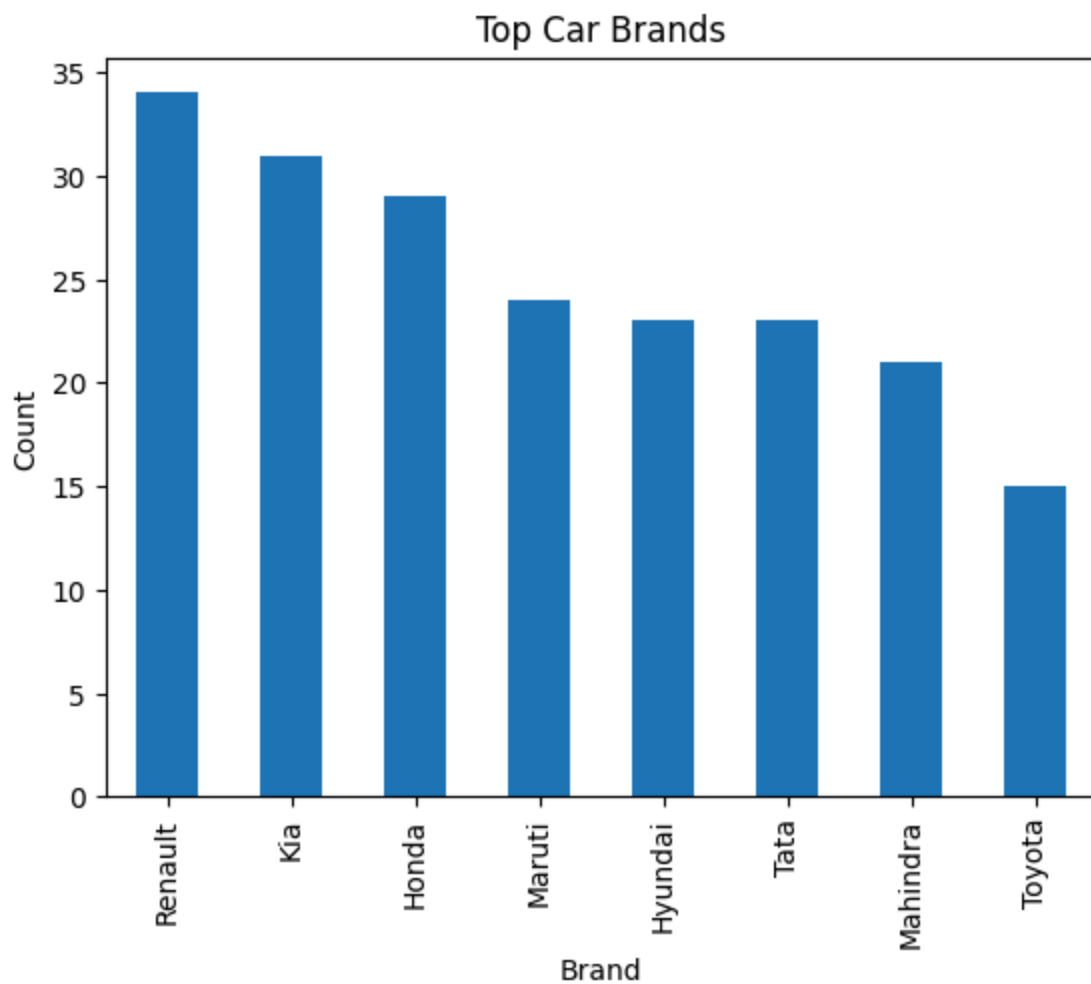
brand	count
Renault	34
Kia	31
Honda	29
Maruti	24
Hyundai	23
Tata	23
Mahindra	21
Toyota	15

dtype: int64

Top Car Brands Bar Chart

```
In [13]: import matplotlib.pyplot as plt

df["brand"].value_counts().head(10).plot(kind="bar")
plt.xlabel("Brand")
plt.ylabel("Count")
plt.title("Top Car Brands")
plt.show()
```



Fuel Type Analysis

What is the distribution of fuel types?

```
In [14]: df["fuel_type"].value_counts()
```

```
Out[14]:
```

	count
--	-------

fuel_type	
-----------	--

Petrol	126
--------	-----

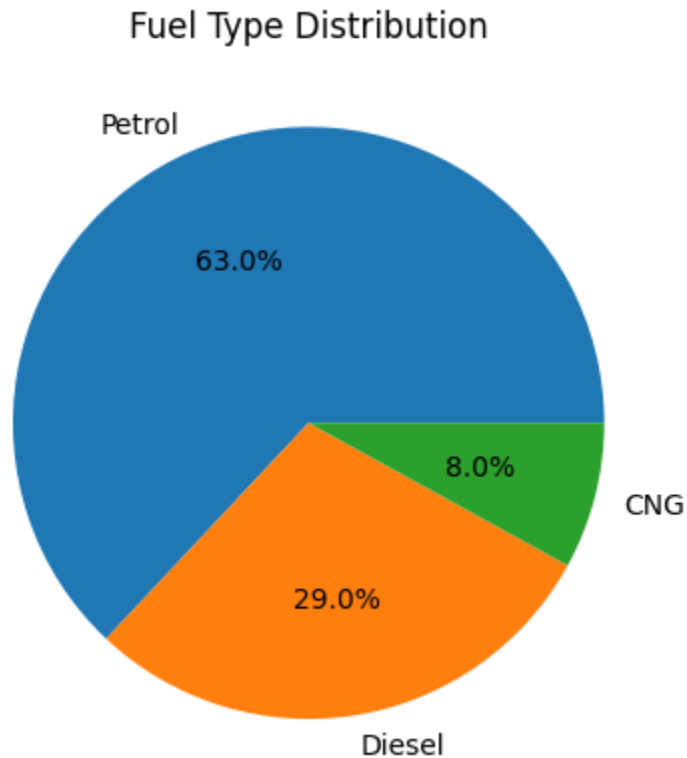
Diesel	58
--------	----

CNG	16
-----	----

dtype: int64

Fuel Type Distribution Pie Chart

```
In [15]: df["fuel_type"].value_counts().plot(kind="pie", autopct="%1.1f%%")
plt.ylabel("")
plt.title("Fuel Type Distribution")
plt.show()
```



Price Type Analysis

What is the average listed price of cars?

```
In [16]: df["listed_price"].mean()
```

```
Out[16]: np.float64(694053.35)
```

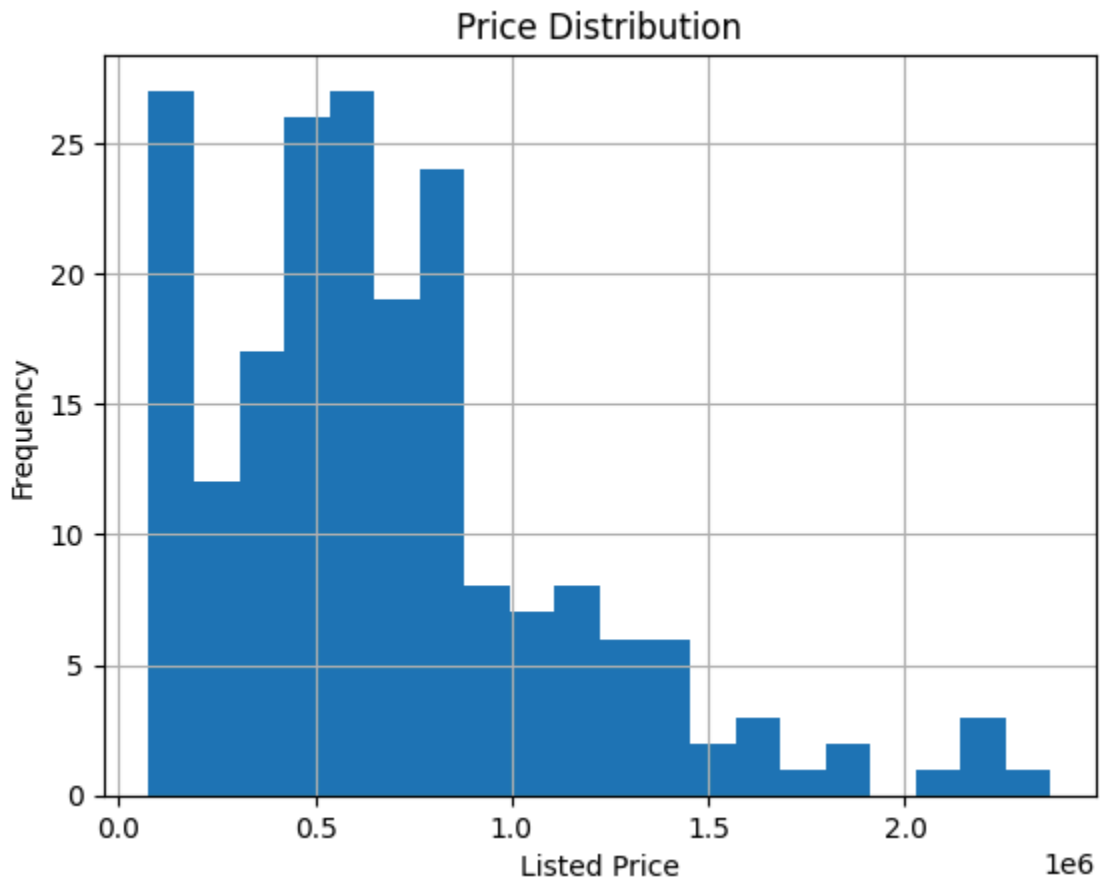
What are the maximum and minimum car prices?

```
In [17]: print(df["listed_price"].max())
print(df["listed_price"].min())
```

```
2370945
76551
```

Car Price Distribution Histogram

```
In [18]: df["listed_price"].hist(bins=20)
plt.xlabel("Listed Price")
plt.ylabel("Frequency")
plt.title("Price Distribution")
plt.show()
```



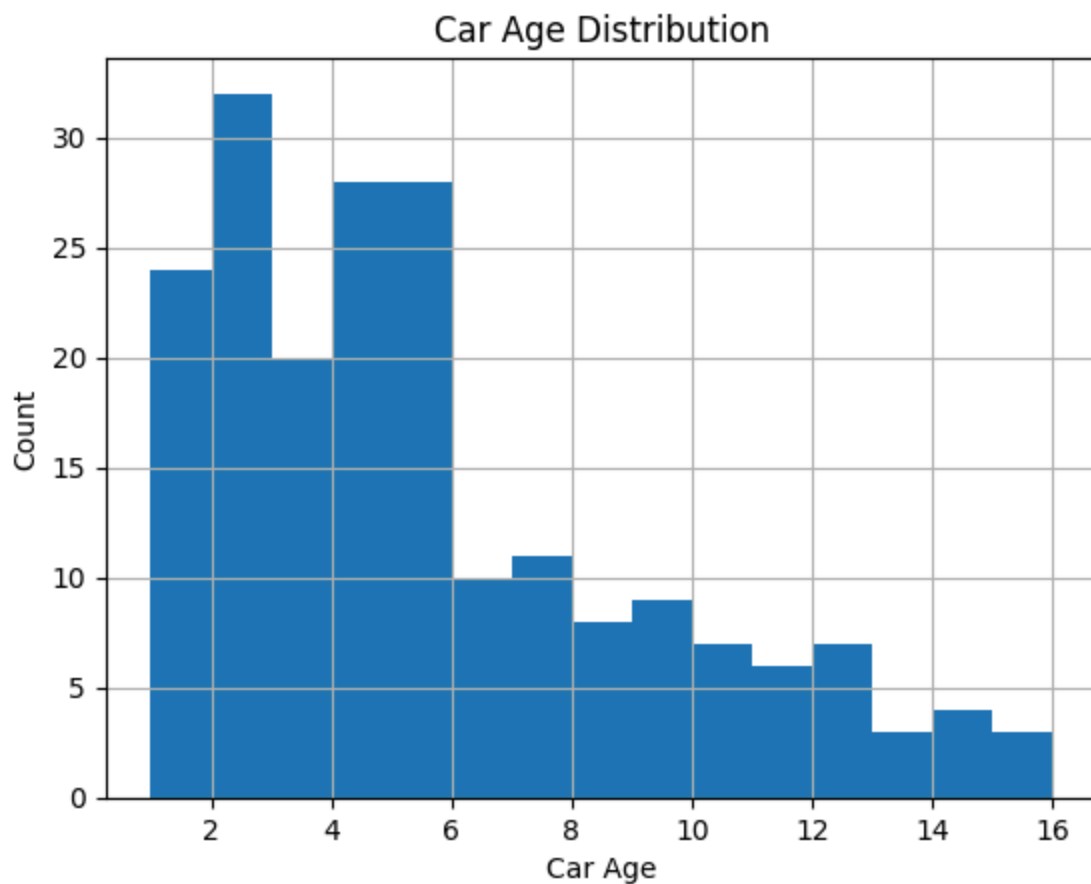
Car Age Analysis

```
In [19]: df["car_age"].mean()
```

```
Out[19]: np.float64(5.215)
```

Car Age Distribution Histogram

```
In [20]: df["car_age"].hist(bins=15)
plt.xlabel("Car Age")
plt.ylabel("Count")
plt.title("Car Age Distribution")
plt.show()
```

Mileage Analysis

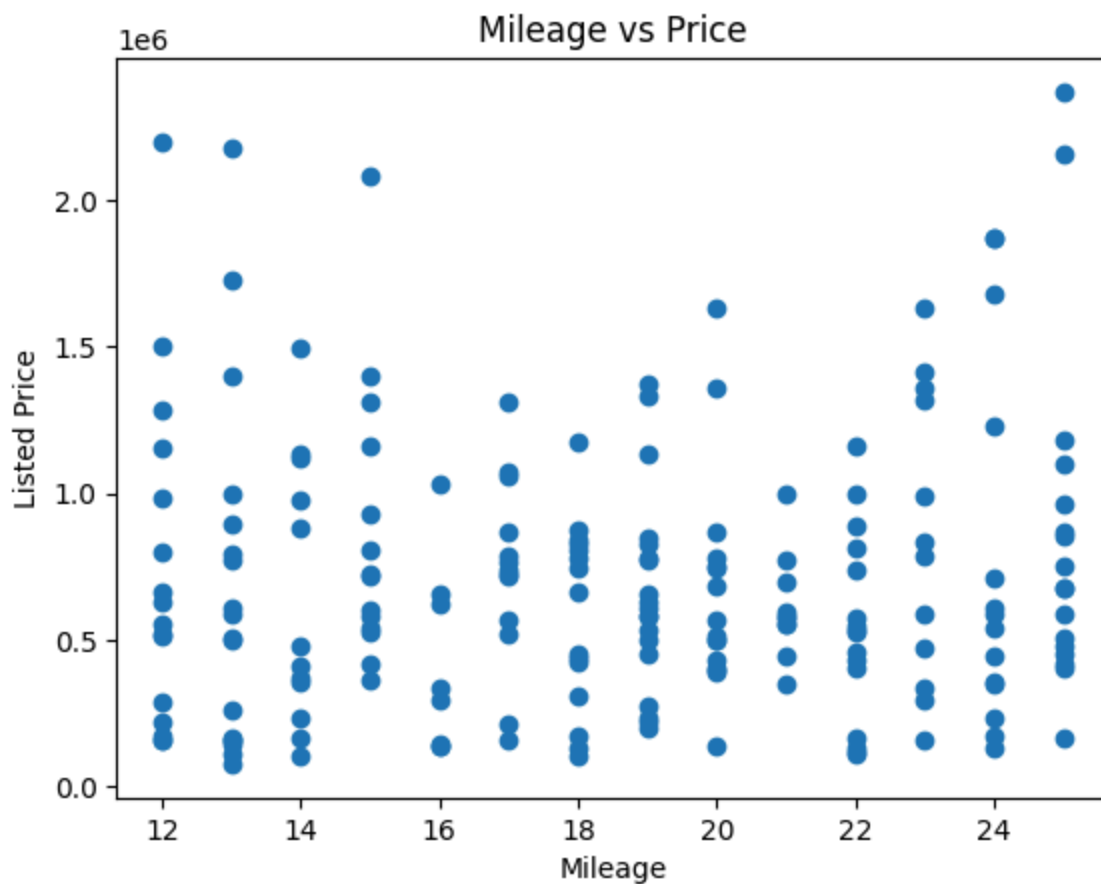
What is the average mileage of cars?

```
In [21]: df["mileage"].mean()
```

```
Out[21]: np.float64(18.515)
```

Mileage vs Listed Price Scatter Plot

```
In [22]: plt.scatter(df["mileage"], df["listed_price"])  
plt.xlabel("Mileage")  
plt.ylabel("Listed Price")  
plt.title("Mileage vs Price")  
plt.show()
```



Transmission Analysis

Which transmission type is more common?

```
In [23]: df["transmission"].value_counts()
```

```
Out[23]:
```

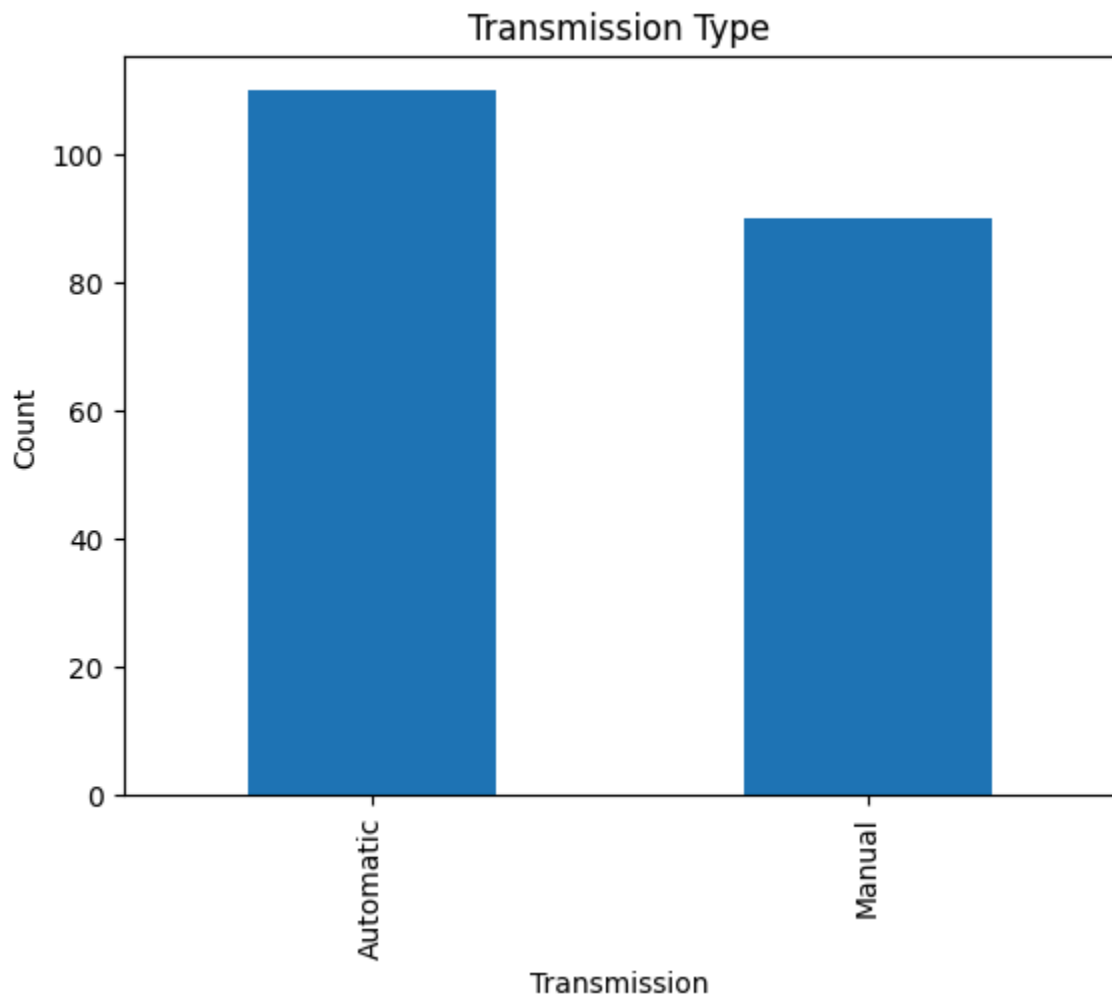
transmission		count
Automatic		110
Manual		90

dtype: int64

Transmission Type Plot Chart

```
In [24]: df["transmission"].value_counts().plot(kind="bar")
plt.xlabel("Transmission")
plt.ylabel("Count")
```

```
plt.title("Transmission Type")  
plt.show()
```



City Analysis

Which city has the highest number of listed cars?

```
In [26]: df["city"].value_counts()
```

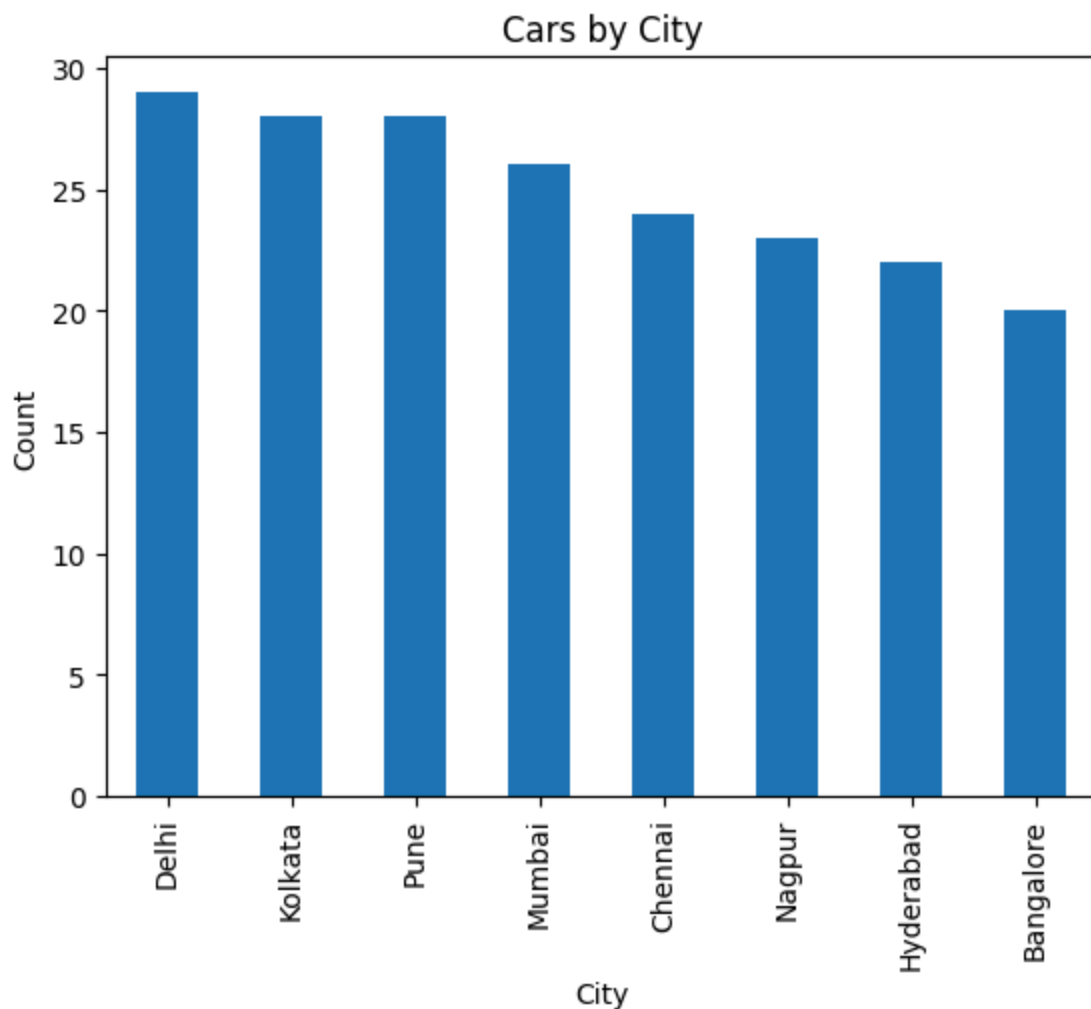
Out[26]:

count	
city	
Delhi	29
Kolkata	28
Pune	28
Mumbai	26
Chennai	24
Nagpur	23
Hyderabad	22
Bangalore	20

dtype: int64

Cars by City Plot Chart

```
In [27]: df["city"].value_counts().plot(kind="bar")
plt.xlabel("City")
plt.ylabel("Count")
plt.title("Cars by City")
plt.show()
```

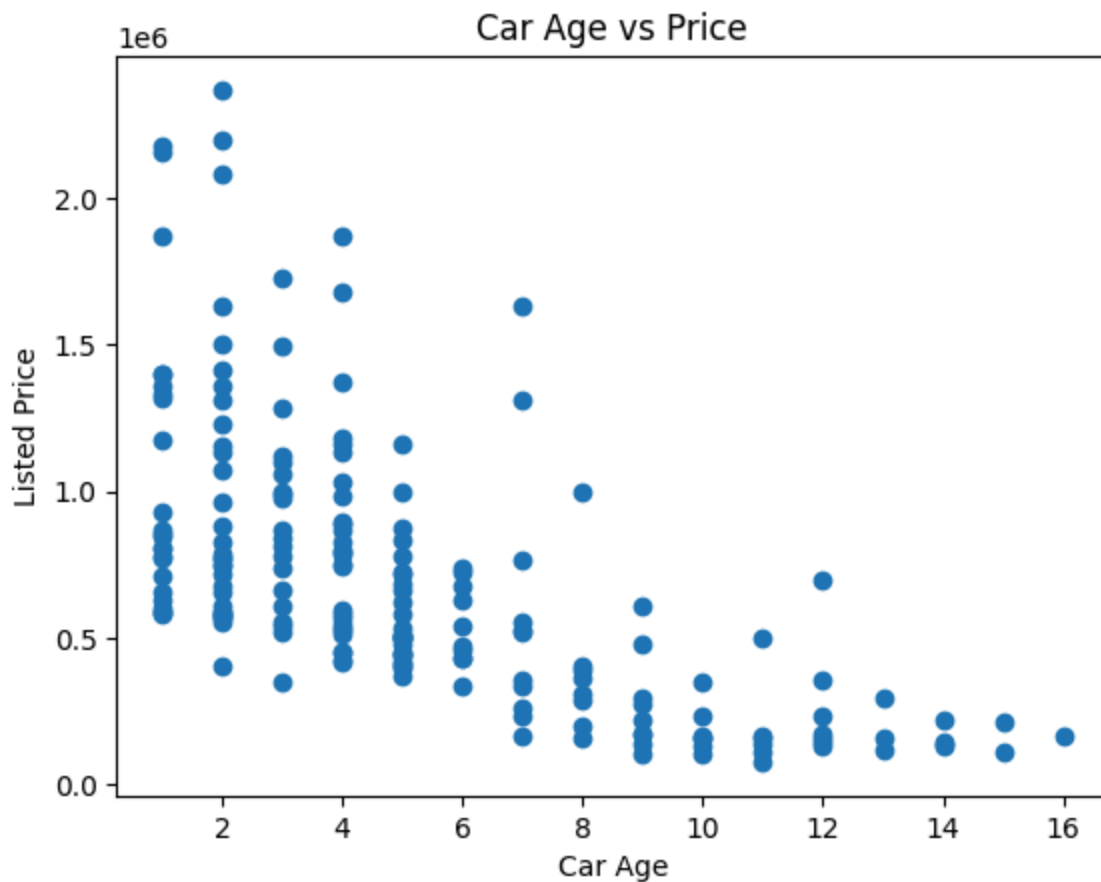


Relationship Analysis

Is there any relationship between car age and listed price?

Car Age vs Listed Price Scatter Plot

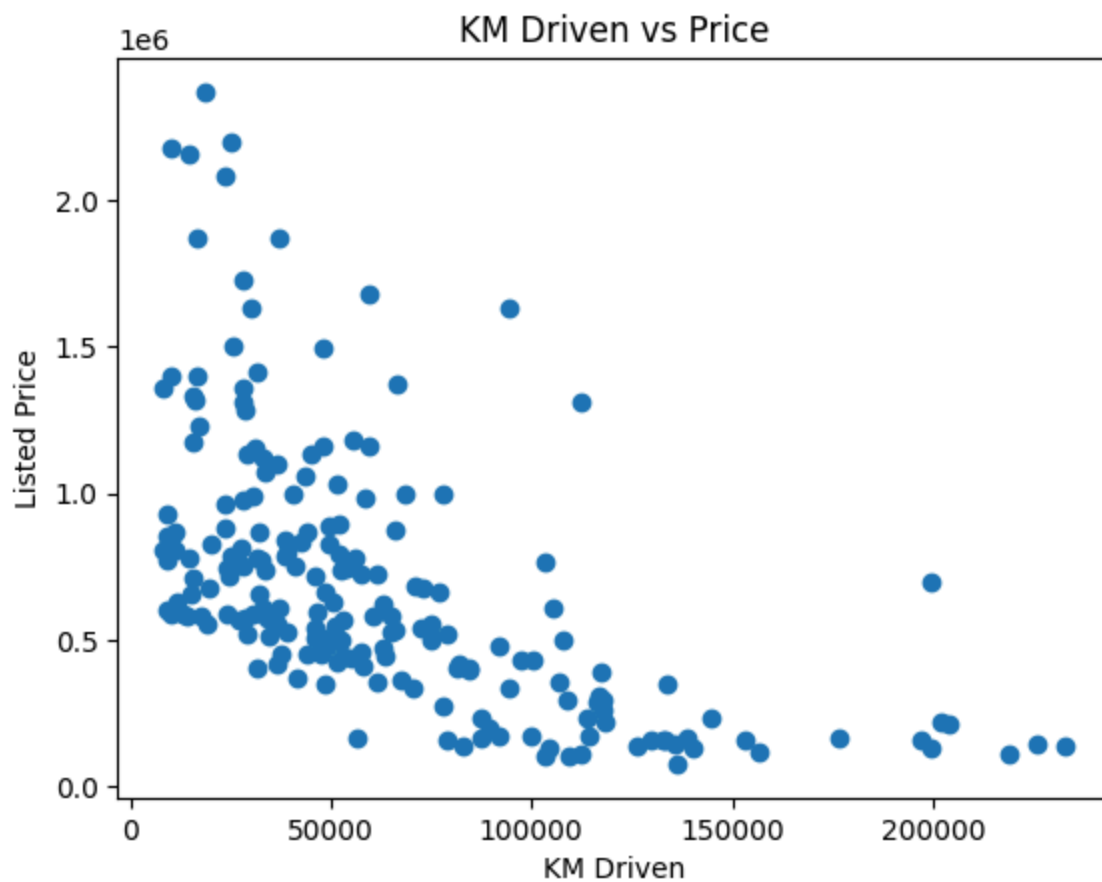
```
In [28]: plt.scatter(df["car_age"], df["listed_price"])
plt.xlabel("Car Age")
plt.ylabel("Listed Price")
plt.title("Car Age vs Price")
plt.show()
```



What is the relationship between kilometers driven and car price?

KM Driven vs Price Scatter Plot

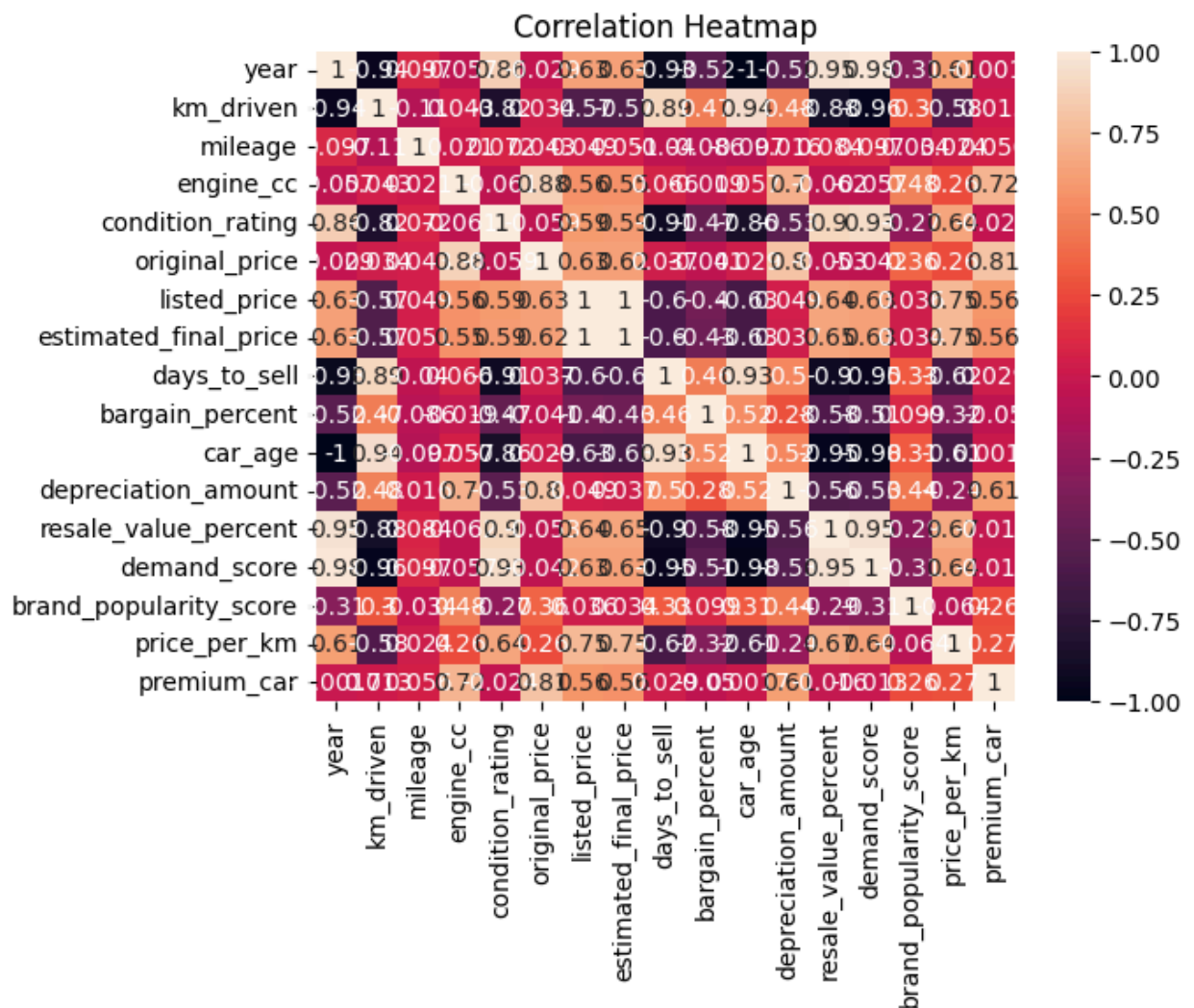
```
In [29]: plt.scatter(df["km_driven"], df["listed_price"])
plt.xlabel("KM Driven")
plt.ylabel("Listed Price")
plt.title("KM Driven vs Price")
plt.show()
```



Correlation Heatmap

```
In [30]: import seaborn as sns

sns.heatmap(df.corr(numeric_only=True), annot=True)
plt.title("Correlation Heatmap")
plt.show()
```



Final Conclusion

This Exploratory Data Analysis (EDA) project analyzed the used car market dataset to understand the relationship between variables like car price, fuel type, transmission, mileage, and kilometers driven. The analysis showed that newer cars with lower mileage usually have higher prices, while older cars with more kilometers driven have lower resale value. Different graphs and visualizations helped identify market trends, customer preferences, and pricing patterns in the used car industry.

Data Visualization

Transmission Type Plot

Cars by City Chart

KM Driven vs Price Scatter Plot

Fuel Type Pie Chart

Car Price Histogram

Car Brand Count Plot

Mileage Distribution Graph

Fuel Type vs Price Graph

Correlation Heatmap